DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III   Topics in Analysis (3TN)

Tutorial Sheet 7: Solutions   **October 2024**

1. Let $(V, \|\cdot\|_V)$ be a Banach space. Let $f : [a,b] \to V$ be a continuous function and $\varepsilon > 0$. If f is differentiable in (a,b) and $\|f'(x) - f'(y)\|_V \le \varepsilon$ for any $x,\, y \in (a,b)$, then prove that

$$\left\| \frac{f(b)-f(a)}{b-a} - f'(t) \right\|_V \le \varepsilon \quad \forall t \in (a,b)\,.$$

[Hint: Consider the function $\phi : [a,b] \to V$ defined by $\phi(s) := f(s) - f(a) - f'(t)(s-a)$ for every $s \in [a,b]$, to which we apply the Lagrange inequality.]

We are indeed in a position to apply the Lagrange inequality as follows: there exists $c \in (a,b)$ such that

$$\boxed{\|\phi(b) - \phi(a)\| \le (b-a)\, \|\phi'(c)\|}\,. \quad (*)$$

Notice that

$$\boxed{\phi(b) = f(b) - f(a) - f'(t)(b-a)\,, \quad \phi(a) = 0\,, \quad \phi'(x) = f'(x) - f'(t)}\,.$$

Hence, we get from $(*)$ that

$$\boxed{\|f(b) - f(a) - f'(t)(b-a)\| \le (b-a)\, \|f'(c) - f'(t)\|}\,.$$

Now, using the inequality

$$\boxed{\|f'(x) - f'(y)\|_V \le \varepsilon \quad \text{for any } x,\, y \in (a,b)}\,,$$

we obtain that

$$\boxed{\|f(b) - f(a) - f'(t)(b-a)\| \le (b-a)\varepsilon \quad \Leftrightarrow \quad \left\| \frac{f(b)-f(a)}{b-a} - f'(t) \right\|_V \le \varepsilon}\,.$$

2. Let $(V, \|\cdot\|)$ be a normed space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$.

   (a) Define what is meant by the statement that $C \subset V$ is a non-empty convex subset of V.

   We recall that the set $C \subset V$ is a non-empty convex subset of V if either C is reduced by un seul point or for every $x, y \in C$, the segment line $[x,y] \subset C$. We also recall that

$$[x,y] := \{z = \lambda x + (1-\lambda) y\colon\ \lambda \in [0,1]\}\,.$$

(b) Show that the open ball $B(x_0, r)$ is a convex subset of V.

Let $x, y \in B(x_0, r)$, we want to show that the segment line $[x, y] \subset B(x_0, r)$. Let $\lambda \in [0, 1]$ and let $z = \lambda x + (1-\lambda)y$. So, we assume that $\lambda \in (0, 1)$.We find that

$$\begin{aligned} \|z - x_0\| &= \|(\lambda x + (1-\lambda)y) - (\lambda x_0 + (1-\lambda)x_0)\| \\ &= \|\lambda(x - x_0) + (1-\lambda)(y - x_0\| \leq \|\lambda(x - x_0)\| + \|(1-\lambda)(y - x_0)\| \\ &= \lambda \|x - x_0\| + (1-\lambda)\|y - x_0\| < \lambda r + (1-\lambda)r = r\,. \end{aligned}$$

Hence, $z \in B(x_0, r)$ which shows that $B(x_0, r)$ is a convex subset of V.

(c) Show that if C is a convex subset of V, then its closure $\overline{C}$ is also closed.

3. Let $(V, +, \cdot)$ be a real vector space. Let $V := C([0, 1], \mathbb{R})$ be the space of continuous functions from $[0, 1]$ to $\mathbb{R}$. Consider the mapping $N : V \to \mathbb{R}$ defined by

$$N(f) := \sup_{x\in[0,1]} e^{\arctan(1-x)}|f(x)|\,.$$

(i) Show that N is a norm on V and that N is equivalent to the norm $\|\cdot\|_\infty$.

- $\forall f \in V$, $N(f) \geq 0$ and

$$\begin{aligned} N(f) = 0 \;\Leftrightarrow\; & \sup_{x\in[0,1]} e^{\arctan(1-x)}|f(x)| \quad\Leftrightarrow\quad e^{\arctan(1-x)}|f(x)| \quad \forall x \in [0, 1] \\ \Leftrightarrow\; & f(x) = 0 \quad \forall x \in [0, 1]\,. \end{aligned}$$

- Let $f \in V$ and let $\lambda \in \mathbb{R}$, we get that

$$N(\lambda f) = \sup_{x\in[0,1]} e^{\arctan(1-x)}|\lambda f(x)| = |\lambda| \sup_{x\in[0,1]} e^{\arctan(1-x)}|f(x)| = |\lambda| N(f)\,.$$

- Let $f, g \in V$, we get that

$$\begin{aligned} N(f + g) &= \sup_{x\in[0,1]} e^{\arctan(1-x)}|f(x) + g(x)| \leq \sup_{x\in[0,1]} e^{\arctan(1-x)}[|f(x)| + |g(x)|] \\ &\leq \sup_{x\in[0,1]} e^{\arctan(1-x)}|f(x)| + \sup_{x\in[0,1]} e^{\arctan(1-x)}|g(x)| = N(f) + N(g)\,. \end{aligned}$$

(ii) Let

$$C := \{f \in V\colon\ \sin x \leq f(x) \leq 2x \quad \forall x \in [0, 1]\}$$

then, show that C is a **non-empty closed convex** subset of (V, N).

Notice that the following functions belong to C:

$$f_1(x) = \sin x, \quad f_2(x) = x, \quad f_3(x) = 2x\,.$$

Hence, $C \neq \emptyset$.

Claim 1: C is convex. Let $f, g \in C$ and let $\lambda \in [0, 1]$. Then we write

$$\begin{cases} f \in C \quad\Leftrightarrow\quad \sin x \leq f(x) \leq 2x \quad x \in [0, 1], \quad (1) \\ g \in C \quad\Leftrightarrow\quad \sin x \leq g(x) \leq 2x \quad x \in [0, 1], \quad (2) \end{cases}$$

So, we obtain the operation $\lambda \times (1) + (1-\lambda) \times (2)$ that for every $x \in [0,1]$

$$\sin x = \lambda \sin x + (1-\lambda)\sin x \leq \lambda f(x) + (1-\lambda) g(x) \leq \lambda(2x) + (1-\lambda)(2x) = 2x\,.$$

Thus, $\lambda f + (1-\lambda) g \in C$.

Claim 2: C is closed in (V, N). Let (f_n) be a sequence in C which converges to f in the norm N, that is,

$$\lim_{n\to\infty} N(f_n - f) = 0\,.$$

We want to show that $f \in C$. Notice that, for every $x \in [0,1]$, we have that $1 \leq e^{\arctan(1-x)}$ and hence,

$$\begin{aligned} 0 \leq |f_n(x) - f(x)| \leq e^{\arctan(1-x)}|f_n(x) - f(x)| &\leq \sup_{t\in[0,1]} e^{\arctan(1-t)}|f_n(t) - f(t)| \\ &= N(f_n - f) \to 0\,. \end{aligned}$$

Therefore, we obtain by the squeeze theorem that $\lim_{n\to\infty} f_n(x) = f(x)$ which means that (f_n) converges pointwise to f.
Now, we write

$$\sin(x) \leq f_n(x) \leq 2x \quad \forall x \in [0,1], \ \ \forall n \in \mathbb{N}\,.$$

Passing to the limit as $n \to \infty$, we obtain that

$$\sin x \leq f(x) \leq 2x, \quad \forall x \in [0,1]\,.$$

which means that $f \in C$.

4. Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give a counterexample.

   (a) Let $(V, +, \cdot)$ be a vector space over $\mathbb{R}$. If C_1 and C_2 are convex subsets of V with $C_1 \cap C_2 \neq \emptyset$, then each set $C_1 \cap C_2$, $C_1 \cup C_2$ and $C_1 + C_2$ is convex.

   - **True!** $C_1 \cap C_2$ is convex: let $u, v \in C_1 \cap C_2$ and let $\lambda \in [0,1]$

$$\begin{cases} u, v \in C_1 & \Rightarrow \quad \lambda u + (1-\lambda) v \in C_1, \\ u, v \in C_2 & \Rightarrow \quad \lambda u + (1-\lambda) v \in C_2, \end{cases} \quad \Rightarrow \quad \lambda u + (1-\lambda) v \in C_1 \cap C_2\,.$$

   - **False!** $C_1 \cup C_2$ is not always convex. In fact, $C_1 = \{0\} \times [0,1] \subset \mathbb{R}^2$ is convex, $C_2 := [0,1] \times \{0\} \subset \mathbb{R}^2$ is convex with $C_1 \cap C_2 = \{(0,0)\}$. But $C_1 \cup C_2$ is not convex because $u = (0,1) \in C_1 \cup C_2$ and $v = (1,0) \in C_1 \cup C_2$ but $w = (1/2, 1/2)$ belongs to $[u, v]$ but $w \notin C_1 \cup C_2$..

   - **True!** $C_1 + C_2$ is convex: let $u, v \in C_1 + C_2$ and let $\lambda \in [0,1]$. So, $u = u_1 + u_2$ and $v = v_1 + v_2$ with $u_1, v_1 \in C_1$ and $u_2, v_2 \in C_2$. Hence,

$$\begin{aligned} &\lambda u + (1-\lambda) v = \lambda(u_1 + u_2) + (1-\lambda)(v_1 + v_2) \\ = \ &\underbrace{[\lambda u_1 + (1-\lambda) v_1]}_{\in C_1 \text{ (because } C_1 \text{ is convex)}} + \underbrace{[\lambda u_2 + (1-\lambda) v_2]}_{\in C_2 \text{ (because } C_2 \text{ is convex)}} \in C_1 + C_2\,. \end{aligned}$$

(b) Let A be a square matrix $n \times n$. Let $b \in \mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) := \phi(Ax + b)$ for some function $\phi : \mathbb{R}^n \to \mathbb{R}^n$. If ϕ is convex, then the function f is also convex.

True! Let $x, y \in \mathbb{R}^n$ and let $\lambda \in [0, 1]$.

$$
\begin{aligned}
& f(\lambda x + (1-\lambda)y) = \phi(A(\lambda x + (1-\lambda)y) + b) = \phi(\lambda(Ax + b) + (1-\lambda)(Ay + b)) \\
\leq \;& \lambda\phi(Ax + b) + (1-\lambda)\phi(Ay + b) = \lambda f(x) + (1-\lambda)f(y)\,.
\end{aligned}
$$

(c) If $f : \mathbb{R}^n \to \mathbb{R}$ is convex, then also λf is convex for every $\lambda \in \mathbb{R}$.

False! $f(x) = x^2$ is convex on $\mathbb{R}$ while $g(x) = -x^2$ is not.